```
apple['SMA20'] = apple['Close'].rolling(window=20).mean()
apple['SMA50'] = apple['Close'].rolling(window=50).mean()

# Generate signals
apple['Signal'] = 0  # 0 means no position
apple['Signal'] = np.where(apple['SMA20'] > apple['SMA50'], 1, 0)  # 1 means buy/hold

# Generate trading orders
apple['Position'] = apple['Signal'].diff()
```

In this code, we:

1. Download three years of Apple stock data
2. Calculate 20-day and 50-day simple moving averages
3. Create a signal column (1 for buy/hold, 0 for no position) based on whether the short-term average is above the long-term average
4. Calculate position changes (1 for buy, -1 for sell, 0 for hold)

While extremely simple, this strategy captures the essence of trend-following, one of the most widely used approaches in algorithmic trading.

*?? **Did You Know?** The legendary Turtle Traders, a group of ordinary people trained by commodity trader Richard Dennis in the 1980s, used a type of moving average system to make over $175 million in just five years. Dennis recruited people with no trading experience to prove that trading could be taught using systematic rules. Their primary strategy—a breakout system with trend-following elements similar to moving average crossovers—turned several complete novices into millionaires. This real-world experiment demonstrated that simple, rule-based strategies can be highly effective when properly implemented and followed with discipline.*